The Credentials Arms Race: Understanding Immigrant Over-Education in the U.S. Labor Market

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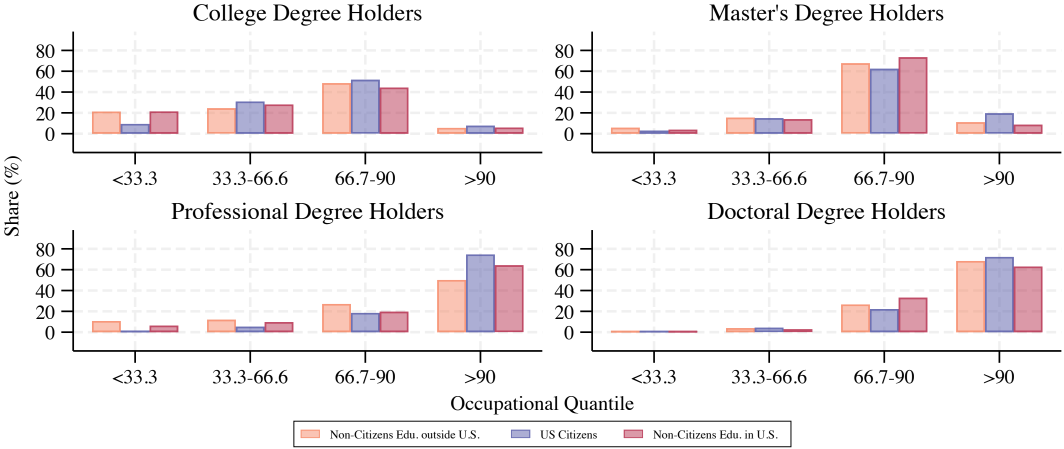
Occupatioal Quantiles

Table 1: Occupational Quantile Examples

Occupation	Doc./Pro.	MA	BA	Assoc.	HS	<HS	Occ. Quantile
Surgeon	99.5	0.5	0	0	0	0	100
Post-secondary Teacher	37.8	33.2	19.4	2.4	6.8	0.4	97.1
Sales Managers	1.0	10.8	50.2	7.5	29.5	1.0	61.0
Respiratory Therapists	0.9	3.1	25.0	56.6	14.1	0.3	59.5
Fence Erectors	0	0.7	3.7	4.4	69.4	21.8	3.0
Earth Drillers, not Oil	0	0.2	4.0	6.0	76.3	13.5	2.4

Motivation

Figure 1: Educational Mismatch of Male Immigrants Aged 25-64 Quartiles



Notes: Data from American Community Survey-5% from 2010 to 2019. Occupational quantile represents the lexicographic order of jobs over the U.S. citizen workers' modal degree and their shares of Doctorate/Professional, Master's, College, Associate's, and High School.

Related Literature

This paper concerns four strands of literature:

1. Wage Penalty and Over-education (Cassidy and Gaulke, 2024; Chiswick et al., 2009, 2013; Hartog, 2000; Kiker et al., 1997; Li et al., 2023; Lu et al., 2021; Verdugo et al., 1989; Warman et al., 2015).

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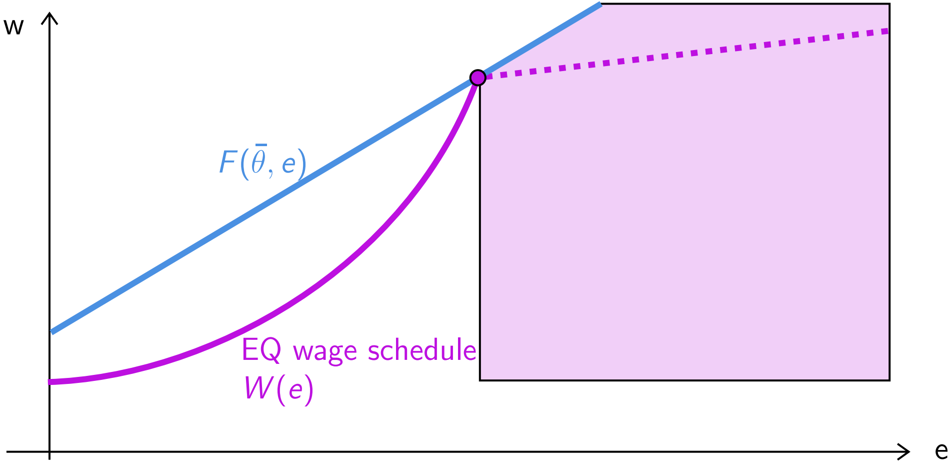
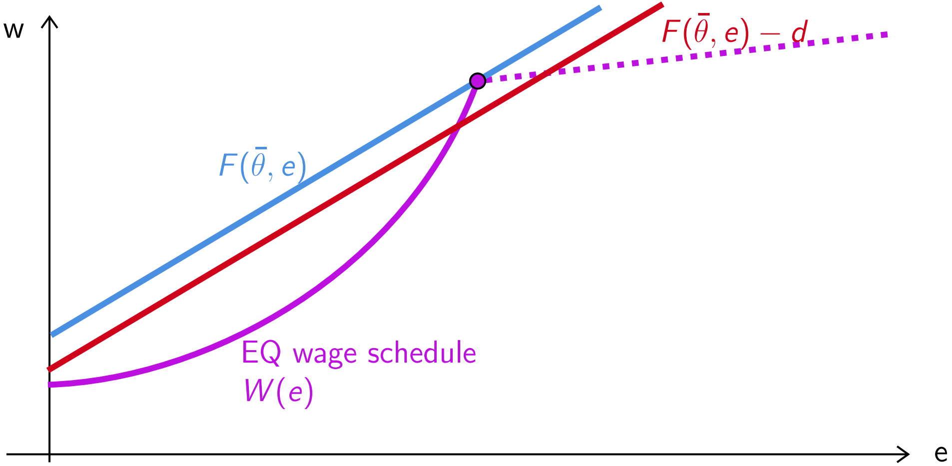
4. Separating equilibria in games with productive signals (Chatterjee et al., 2021; Cho et al., 1987; Kaymak, 2025; Mailath, 1987; Spence, 1978; Tyler et al., 2000).

Continuous Types $\Theta = [\underline{\theta}, \bar{\theta}]$



Continuous Types $\Theta = [\underline{\theta}, \bar{\theta}]$ with New Agents and Costly Hiring



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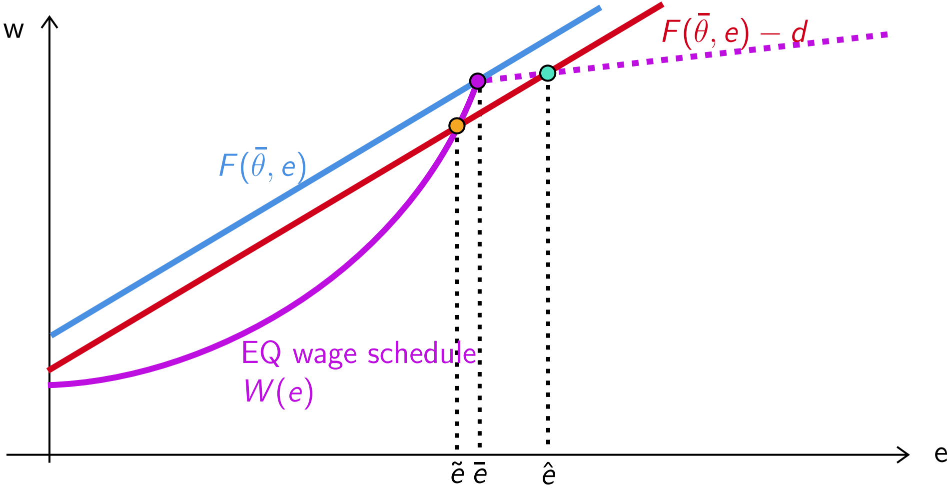

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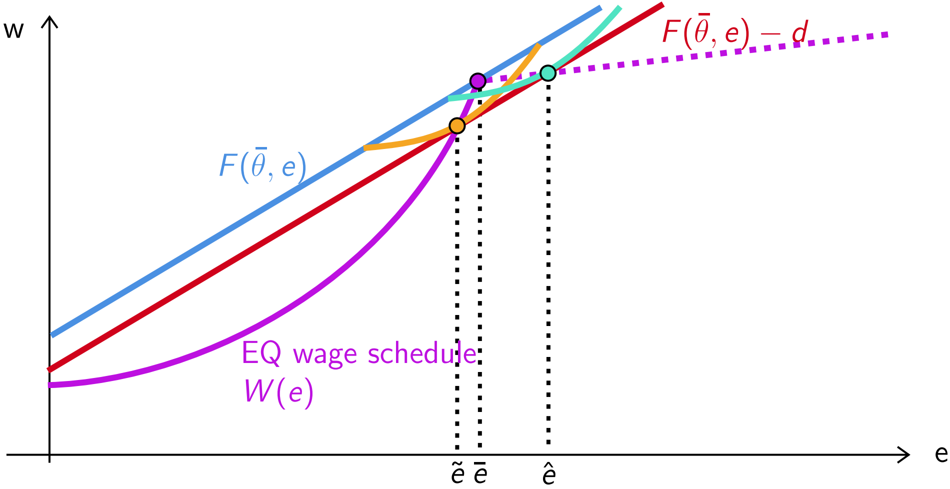

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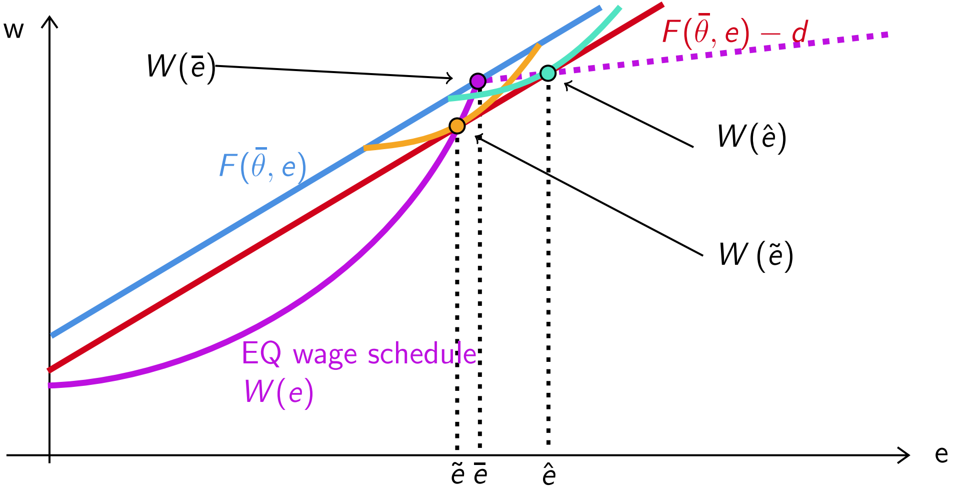

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- Some jobs are less desirable, limiting employers' market power and weakening the need for signaling. (Chiswick et al., 2009; Sattinger, 2012)
  - Some jobs may rely on qualifications other than formal education (Bacon et al., 2020; Chiswick et al., 2008) .

Controlling for Job Characteristics

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2. Importance of Certification vs. Typical On-the-Job Training Graphs

LinkedIn Profiles from People Data Labs

To study whether employer learning reduces education–occupation gaps—something cross-sectional data (e.g., ACS) cannot address—I construct a panel using scraped LinkedIn profiles from People Data Labs.

Current data include:

- Person record (sex, job, location)
- Job experience (dates, company, title, role)
- Education (dates, school, degree, major)
- Certifications (dates)

LinkedIn Panel

I construct a panel of individuals who attended a U.S. degree-granting post-secondary institution and held at least one job matched to an O*NET SOC code. The sample covers 1970–2025 and includes $\sim$44M person-year observations from $\sim$74M individuals, averaging 33 years per person.

Table 2: Combined Education and Employment Statistics

Part 1: By Highest Degree			**Part 2: Education-to-Employment Gap**		
Degree	**Dist. %**	**Jobs**	**Gap Duration**	**Count (N)**	**Pct. %**
Some College	7.53	1.9	Work before (< 0)	28,437,815	64.45
Bachelors	52.50	2.8	Same year (0)	5,314,838	12.05
Masters	31.30	2.6	1 year	2,649,947	6.01
Professional	5.00	2.6	2 years	1,234,768	2.80
PhD	3.70	2.5	3–5 years	2,170,394	4.92
			$>$ 5 years	4,316,303	9.78

Gap Stats: Mean: -3.01, Median: -3.0, SD: 8.81, Total N: 44,124,065

Matching LinkedIn Job Titles to SOCs

Using 250 distinct job titles for each of the 1,016 SOC codes from LinkUp, I fine-tune a Llama-3.1 (8B) model to classify each self-reported job title into its corresponding SOC code.

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> *Input: Vice President, company_name*
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> *Response: 00-0000.00*

Matching LinkedIn Titles to SOCs

Table 3: Example of Predicted SOC Codes using Preliminarily Fine-Tuned Llama-3.1

LinkedIn Title	O*NET SOC	Corresponding O*NET Title
director - field support	11-9151.00	Social and Community Service Managers
senior page designer and copy editor	27-3043.00	Writers and Authors
gucci program for scholars	15-1255.00	NA
taylor mattis fellow	51-9195.04	Glass Blowers, Molders, Benders, and Finishers
project management and marketing	15-1299.09	Information Technology Project Managers
solar analyst intern	47-2151.04	NA
international visiting attorney	23-1011.00	Lawyers
director of lower school admission	11-9033.00	Education Administrators, Postsecondary
asset & reliability engineer	17-2112.00	Industrial Engineers
sales / acct manager	11-2022.00	Sales Managers
attest intern	13-1041.04	Government Property Inspectors and Investigators
immersion associate	25-9044.00	Teaching Assistants, Postsecondary
medicaid waiver case manager	11-9151.00	Social and Community Service Managers

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gfliwn askdm	00-0000.00	NA
taylor mattis fellow	51-9195.05	Potters, Manufacturing
solar analyst intern	17-2199.11	Solar Energy Systems Engineers
international visiting attorney	23-1011.00	Lawyers
director of lower school admission	11-9032.00	Education Administrators, Kindergarten through Secondary
asset & reliability engineer	17-2199.00	NA
sales / acct manager	41-1011.00	First-Line Supervisors of Retail Sales Workers
attest intern	13-2099.00	NA
immersion associate	25-1194.00	Career/Technical Education Teachers, Postsecondary
medicaid waiver case manager	11-9151.00	Social and Community Service Managers
網頁工程師	15-1299.08	Computer Systems Engineers/Architects
博士生	25-9044.00	Teaching Assistants, Postsecondary

Constructing Smooth Education Attainment

With the LinkedIn data, this issue can be solved at each degree level by indexing the schools from which each worker acquired their degree.

$$\text{Score} = 0.65 \times \text{Average Scaled Test Scores} + 0.35 \times \text{Rejection Rate}$$

At each degree level, I can then apply the calculated score to proxy for the signaling value of the degree.

$$OQ_i = \beta_0 + \sum_{d=\text{Mas.}}^{\text{Pro.}} \delta_d \cdot \mathbb{1}\{Deg_i = d\} + \sum_{d=\text{Bac.}}^{\text{Pro.}} \beta_d \cdot \mathbb{1}\{Deg_i = d\} \cdot (score_i - sc\bar{o}re_d) + \varepsilon$$

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$$\hat{e}_i = 1 + \sum_{d=\text{Mas.}}^{\text{Pro.}} \hat{\delta}_d \cdot \mathbb{1}\{Deg_i = d\} + \sum_{d=\text{Bac.}}^{\text{Pro.}} \hat{\beta}_d \cdot \mathbb{1}\{Deg_i = d\} \cdot (score_i - \bar{score}_d)$$

Estimated Education Signals

Table 4: Estimated Marginal Signaling Value of Degrees

Tier	School	Bachelors	Masters	PhD	Professional
2	Bottom 25%	-0.84	2.08	5.39	8.83
3	Top 75%	-0.75	2.06	5.76	9.29
4	Top 50%	-0.62	2.24	5.73	9.05
5	Top 33%	-0.60	1.48	5.70	8.25
6	Top 20%	-0.33	1.52	4.98	8.07
7	Top 10%	-	1.37	4.82	7.61
8	Top 5%	1.18	1.62	4.81	7.55
9	Top 1%	1.64	2.33	4.91	6.78

Future Steps

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- Estimate the cost gap d and the proportion of $\bar{\theta}$ productivity immigrants at ẽ and ê.
- Counterfactual analysis with increases in d—The new $100,000 H-1B fee.
  - May be able to look at which H-1B jobs on the wage schedule became (un)available after policy takes effect in the 2025-2026 cycle.

Questions?

Thank You!

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Population Moments

Table A5: Summary Statistics of Workers Age 25-64 in the US, 2011-2019

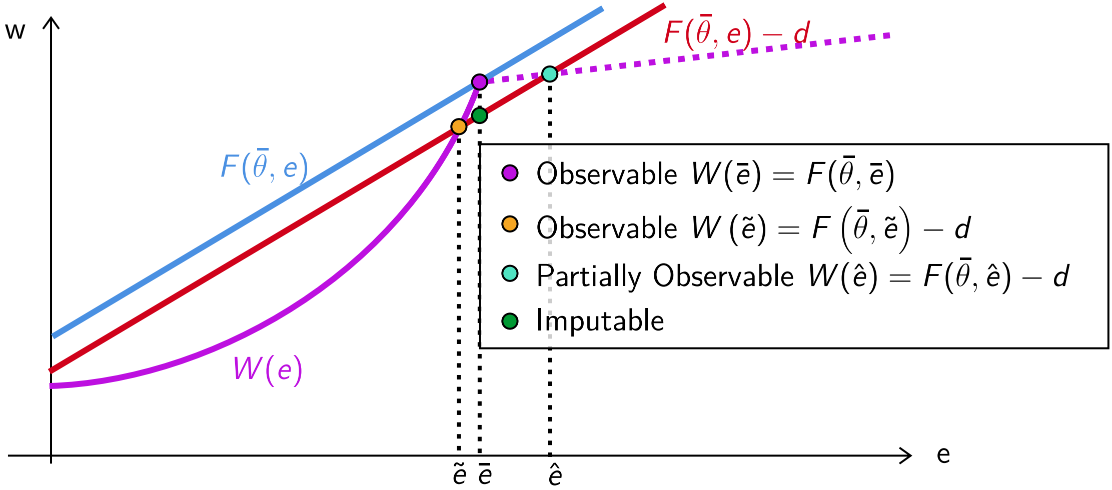
Highest Degree		High School			Associate's			College	
Citizenship	Citizen	Nat. Citizen	Non-citizen	Citizen	Nat. Citizen	Non-citizen	Citizen	Nat. Citizen	Non-citizen
% Workforce	19.9	1.8	2.3	8.8	0.8	0.4	21.2	2.3	1.4
% Single	25	14.8	28.3	21.2	15.1	21.3	26	17.3	21.1
% Married	54.8	67.9	58.3	60.1	67.6	62.5	62	69.9	68.6
Spouse's Citizenship									
% Citizen	59.4	15.2	11.4	63.3	20	20.9	64.1	20.8	16.5
% Nat. Citizen	1.3	38	10.6	1.8	38.6	13	2.3	41.1	10.7
% Non-citizen	1.2	15.4	38.9	1.2	10.4	30.1	1.3	9.1	40.5
Birthplace Official Language									
% Not English		80.1	87.5		74.8	74.5		72.1	64.3
% English		18.4	11.5		23.1	22.7		26.1	33.6
% Mixed		1.4	1.1		2.1	2.9		1.8	2
Median Value									
Age	46	47	39	44	46	41	42	45	39
Years in US		24	14		25	14		23	9
Occ. Quantile	32.5	32.4	22.4	51	47.3	40.5	73	71.9	65.2

Population Moments

Table A6: Summary Statistics of Workers Age 25-64 in the US, 2010-2019

Highest Degree	Master's			Professional			Doctorate		
Citizenship	Citizen	Nat. Citizen	Non-citizen	Citizen	Nat. Citizen	Non-citizen	Citizen	Nat. Citizen	Non-citizen
% Workforce	8.5	1.1	0.8	2.2	0.3	0.2	1.2	0.3	0.2
% Single	19.8	12.1	20.5	18.2	13.9	19.9	17.3	9.7	18.5
% Married	68.1	76.2	73.6	71.4	76.1	71.5	71.9	81.2	76.1
Spouse's Citizenship									
Citizen	67.8	22.4	13.1	69.9	26	15	68.8	22.5	13.1
Nat. Citizen	2.9	46.3	7.9	4	44.5	9.8	4.5	50.7	6.4
Non-citizen	1.5	8	51.3	1.6	6.5	45.4	2.4	7.4	54.1
Birthplace Official Language									
% Not English		62	50.2		63.8	59.2		68	71.2
% English		35.8	48		34.2	39.4		29.6	27.4
% Mixed		2.3	1.8		2	1.4		2.5	1.4
Median Value									
Age	44	46	37	45	47	39	46	49	39
Years in US		23	8		25	8		24	10
Occ. Quantile	83	80	79	98.9	98.9	95.6	97.1	96.8	96.8

New Separating Equilibrium Back



General Separating Equilibrium

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In a full **separating** equilibrium, σ is a one-to-one function, i.e.,

$$\widehat{\Theta}(e) = \{\sigma^{-1}(e)\}, \forall e \in \mathcal{E} \tag{1}$$

and $\sigma(\theta)$ must satisfy *strict incentive compatibility*:

$$\{\sigma(\theta)\} = \underset{e}{argmax}\ U(\theta, \sigma^{-1}(e), e), \forall \theta \in \Theta \tag{2}$$

Regularity Conditions

1. (Smoothness) $U(\theta, \hat{\theta}, e)$ is twice differentiable on $\Theta^2 \times \mathcal{E}$.
2. (Belief Monotonicity) $\dfrac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right) > 0$.
3. (Type monotonicity) $\dfrac{\partial^2}{\partial \theta \partial e} U\left(\theta, \hat{\theta}, e\right) > 0$
4. ("Strict" Quasi-Concavity) $\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right) = 0$ has a unique solution in e, denoted $\phi(\theta)$ such that $\phi(\theta) \in \underset{e}{argmax}\ U\left(\theta, \hat{\theta}, e\right)$, and $\dfrac{\partial^2}{\partial e^2} U\left(\theta, \hat{\theta}, e\right)\Big|_{e=\phi(\theta)} < 0$.
5. (Boundedness) $\exists k > 0$ such that

$$\forall (\theta, e) \in \Theta \times \mathcal{E}, \frac{\partial^2}{\partial e^2} U(\theta, \hat{\theta}, e) \geq 0 \ \Rightarrow \ \left| \frac{\partial}{\partial e} U(\theta, \hat{\theta}, e) \right| > k.$$

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2. (Belief Monotonicity) $\frac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right) > 0$.
3. (Type monotonicity) $\frac{\partial^2}{\partial \theta \partial e} U\left(\theta, \hat{\theta}, e\right) > 0$
4. ("Strict" Quasi-Concavity) $\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right) = 0$ has a unique solution in e, denoted $\phi(\theta)$ such that $\phi(\theta) \in \underset{e}{argmax}\ U\left(\theta, \hat{\theta}, e\right)$, and $\left.\frac{\partial^2}{\partial e^2} U\left(\theta, \hat{\theta}, e\right)\right|_{e=\phi(\theta)} < 0$.
5. (Boundedness) $\exists k > 0$ such that

$$\forall (\theta, e) \in \Theta \times \mathcal{E}, \frac{\partial^2}{\partial e^2} U(\theta, \hat{\theta}, e) \geq 0 \ \Rightarrow \ \left| \frac{\partial}{\partial e} U(\theta, \hat{\theta}, e) \right| > k.$$

Additionally, there are two niceness conditions: Initial Value (IV) and Single Crossing (SC).

Characterization of the Solution

Mailath (1987) shows that
If conditions 1-5 are satisfied, then a strictly increasing function σ, continuous at $\underline{\theta}$, satisfies strict incentive compatibility if and only if:

1. σ solves $$\frac{d\sigma}{d\theta} = -\frac{\frac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right)}{\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right)}$$
2. when $\frac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right) > 0$, $\dfrac{\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right)}{\frac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right)}$ is a strictly increasing function of θ.

Moreover, the strictly increasing σ that satisfies strict incentive compatibility is unique.

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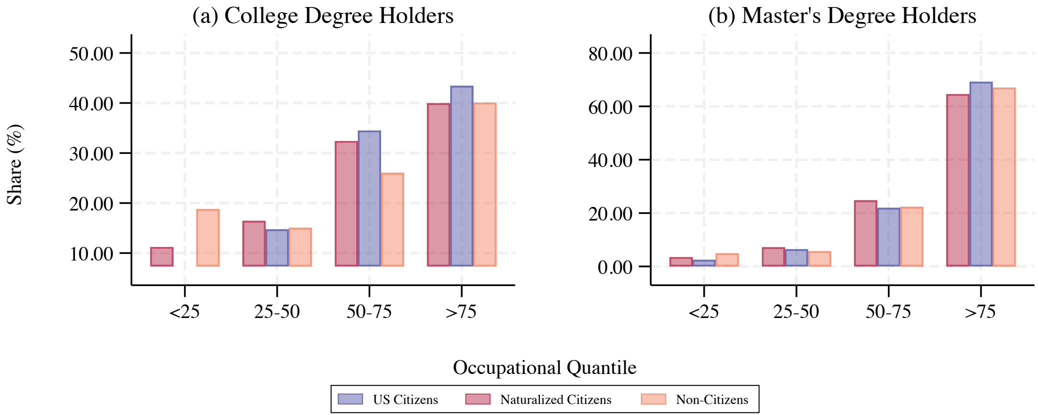
2. when $\frac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right) > 0$, $\frac{\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right)}{\frac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right)}$ is a strictly increasing function of θ.

Moreover, the strictly increasing σ that satisfies strict incentive compatibility is unique.

In other words, there is an unique full separating equilibrium in the natives' labor market given the assumptions in the basic model environment.

Figure A2: Educational Mismatch of Male Immigrants at High Degree Levels, 2010-2019



Notes: Data from American Community Survey-5% from 2010 to 2019. Occupational quantile represents the lexicographic order of jobs over the shares of U.S. citizen workers' education attainment of Doctorate/Professional, Master's, College, Associate's, and High School.

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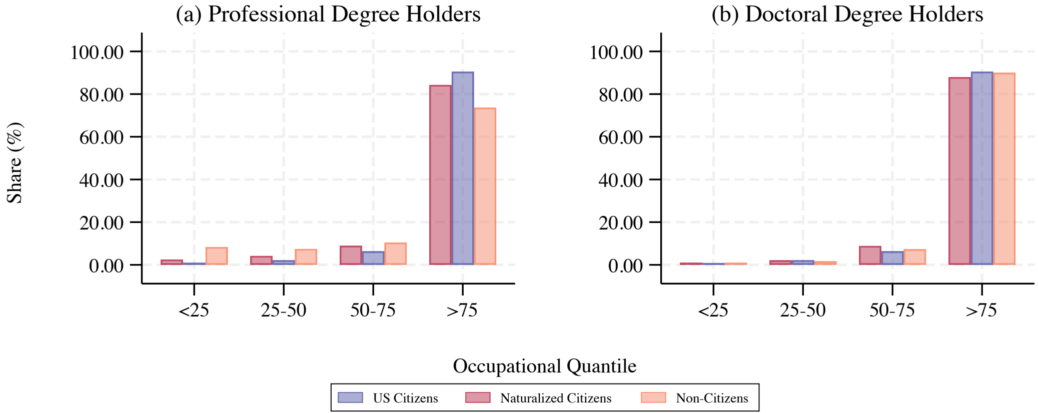
Figure A3: Educational Mismatch of Male Immigrants at High Degree Levels, 2010-2019



Notes: Data from American Community Survey-5% from 2010 to 2019. Occupational quantile represents the lexicographic order of jobs over the shares of U.S. citizen workers' education attainment of Doctorate/Professional, Master's, College, Associate's, and High School.

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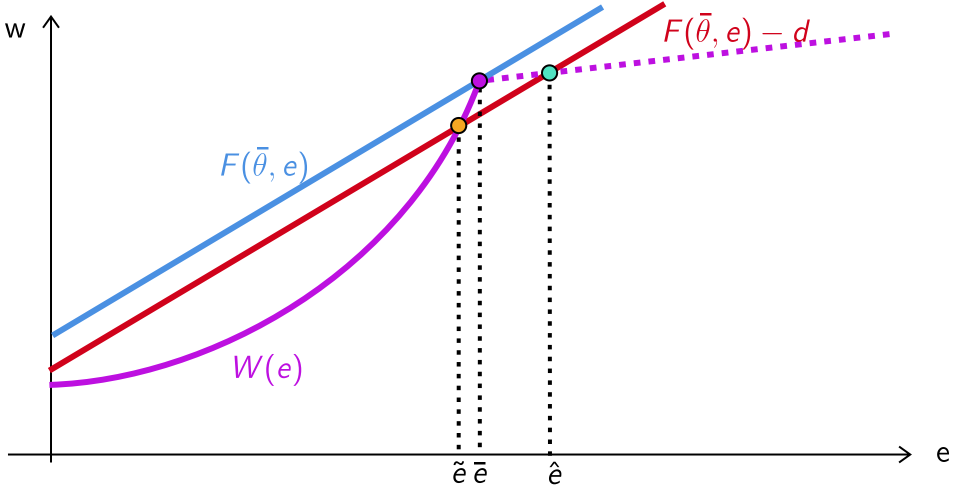
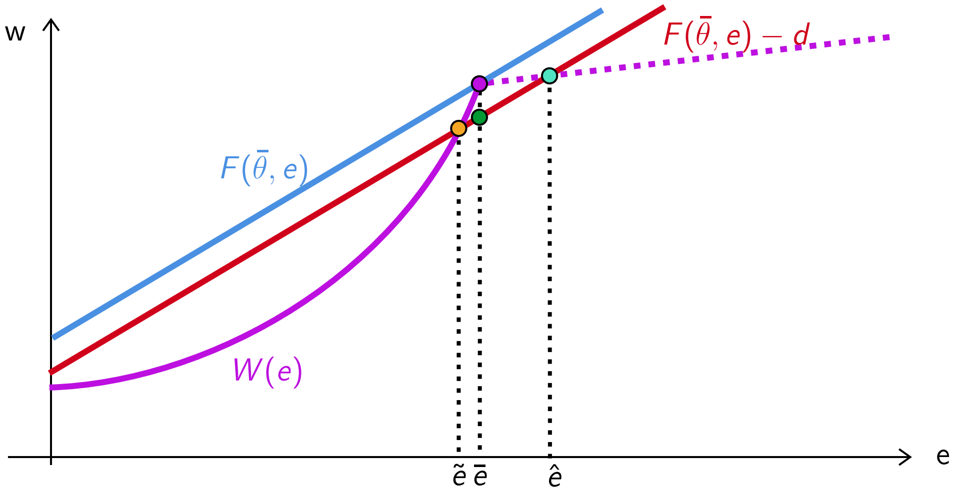
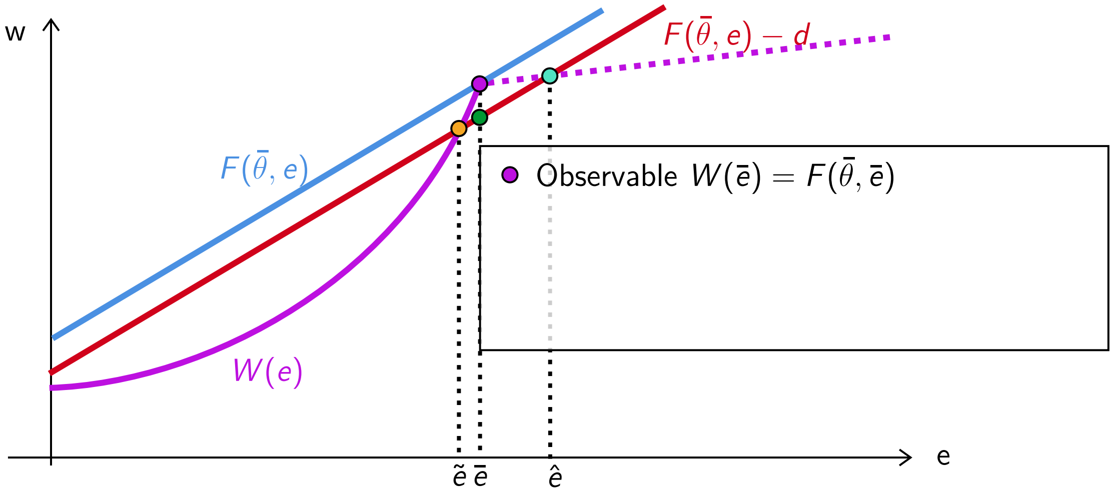
New Separating Equilibrium



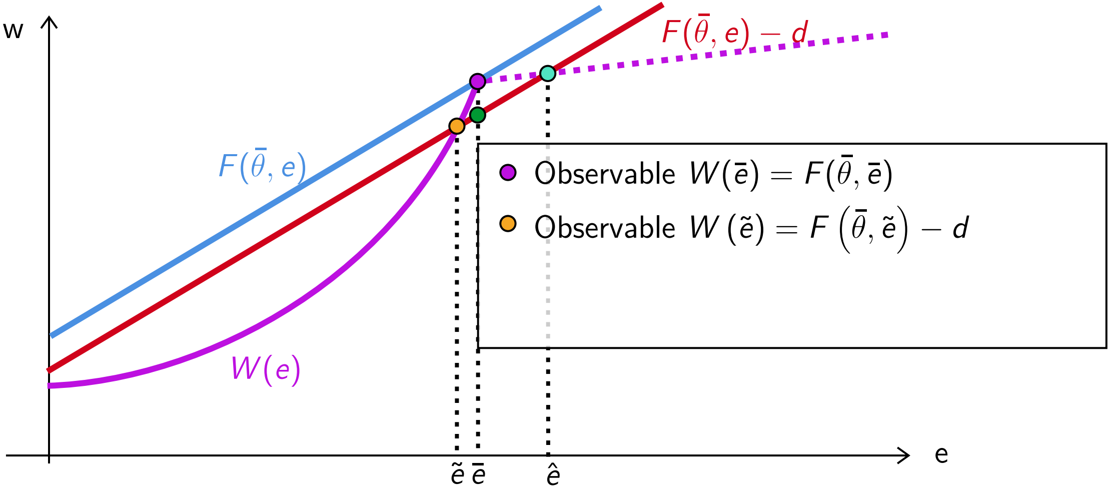
New Separating Equilibrium



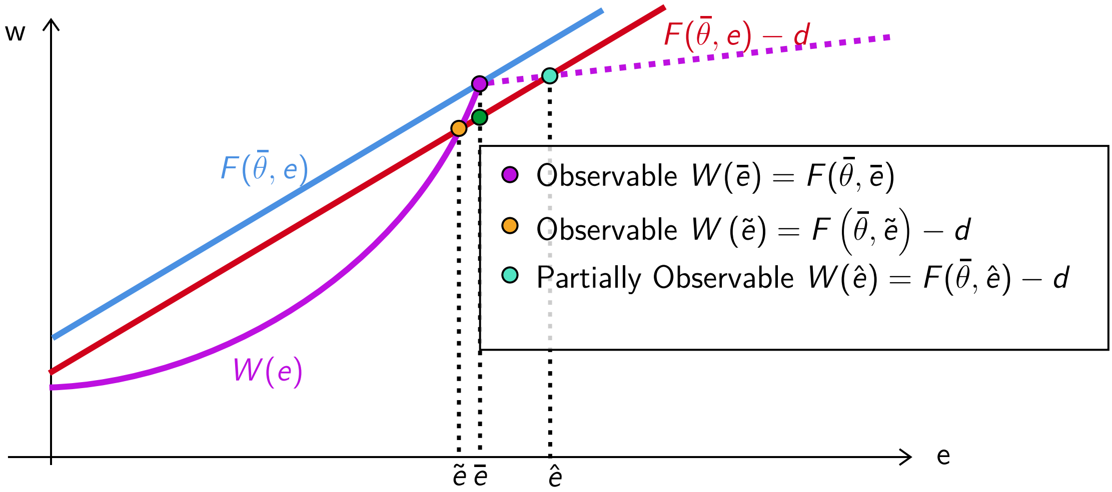
New Separating Equilibrium



New Separating Equilibrium



New Separating Equilibrium



New Separating Equilibrium

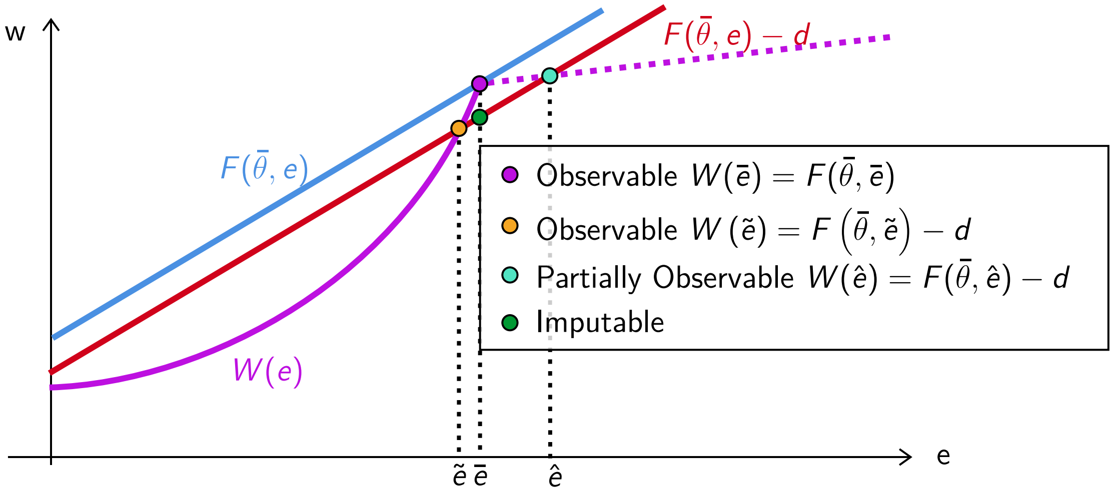
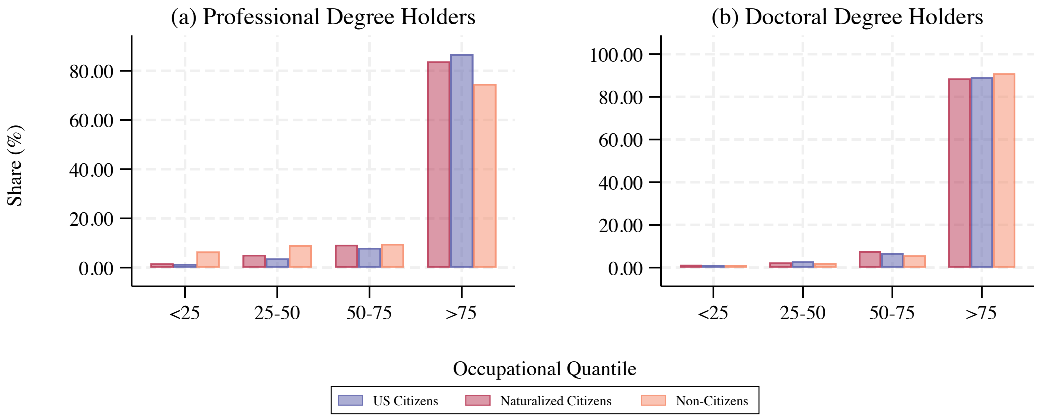


Figure A4: Educational Mismatch of Single Immigrants at High Degree Levels, 2010-2019



Notes: Data from American Community Survey-5% from 2010 to 2019. Occupational quantile represents the lexicographic order of jobs over the shares of U.S. citizen workers' education attainment of Doctorate/Professional, Master's, College, Associate's, and High School.

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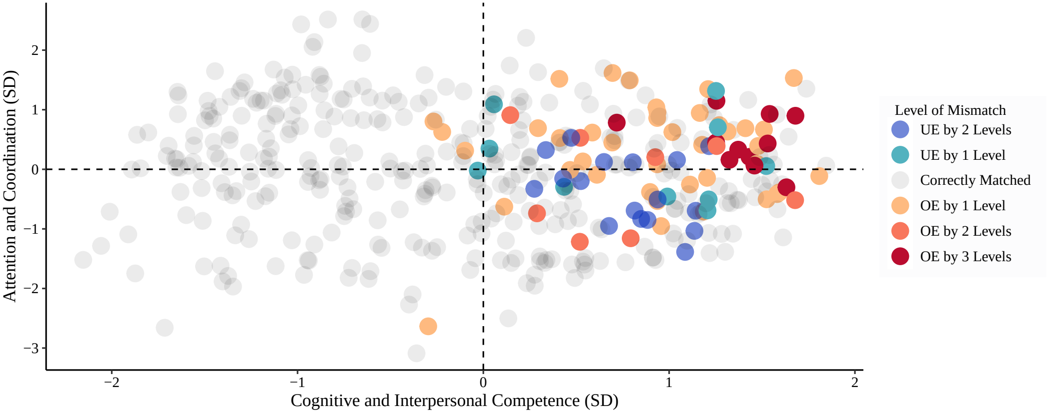
Figure A5: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US



Notes: Level of mismatch is the difference between the modal degree of a non−citizen worker, whose highest degree is obtained inside the United States, and the modal degree of a citizen worker. The degree levels, from highest to lowest, are: Doctorate, Professional, Master's, Bachelor's, Associates, High School, and less than High School.

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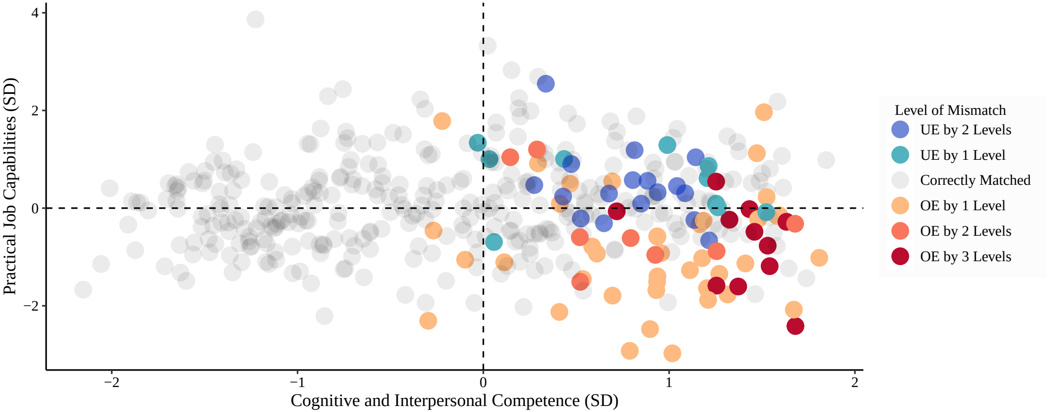
Figure A6: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US



Notes: Level of mismatch is the difference between the modal degree of a non−citizen worker, whose highest degree is obtained inside the United States, and the modal degree of a citizen worker. The degree levels, from highest to lowest, are: Doctorate, Professional, Master's, Bachelor's, Associates, High School, and less than High School.

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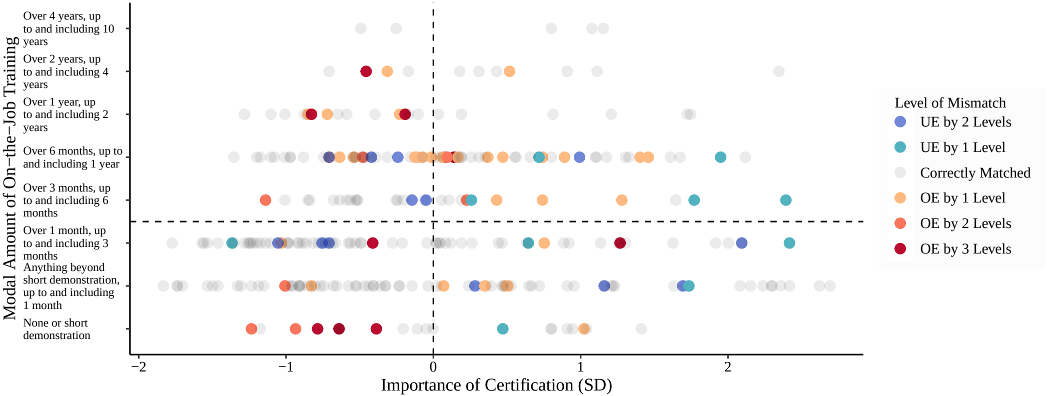
Figure A7: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US



Notes: Level of mismatch is the difference between the modal degree of a non−citizen worker, whose highest degree is obtained inside the United States, and the modal degree of a citizen worker. The degree levels, from highest to lowest, are: Doctorate, Professional, Master's, Bachelor's, Associates, High School, and less than High School. Job characteristics are from the O*NET dataset. Mismatch realizations are from the American Community Survey 2010−2019.

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Location of Highest Degree

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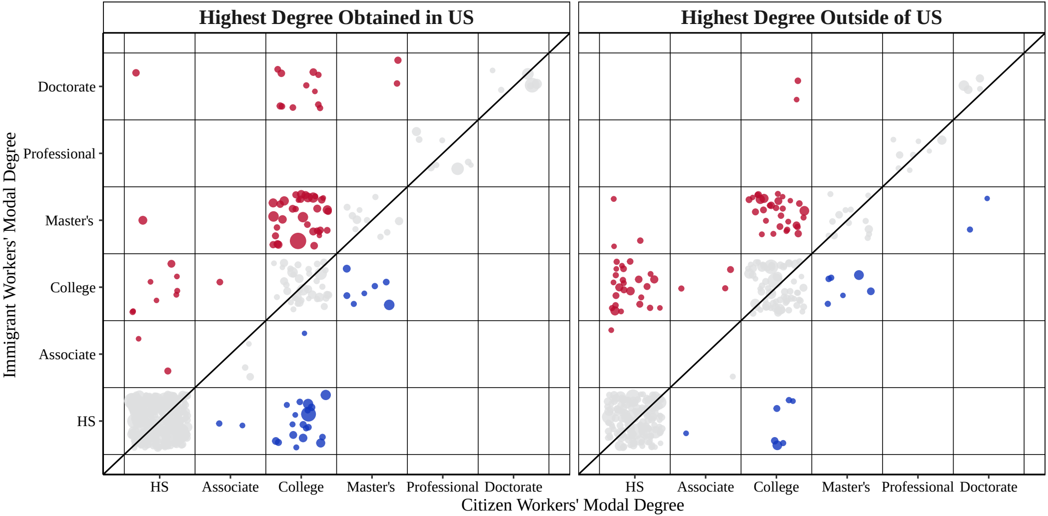
Figure A8: Educational Mismatch of Non-Citizens, 2010-2019



Location of Highest Degree

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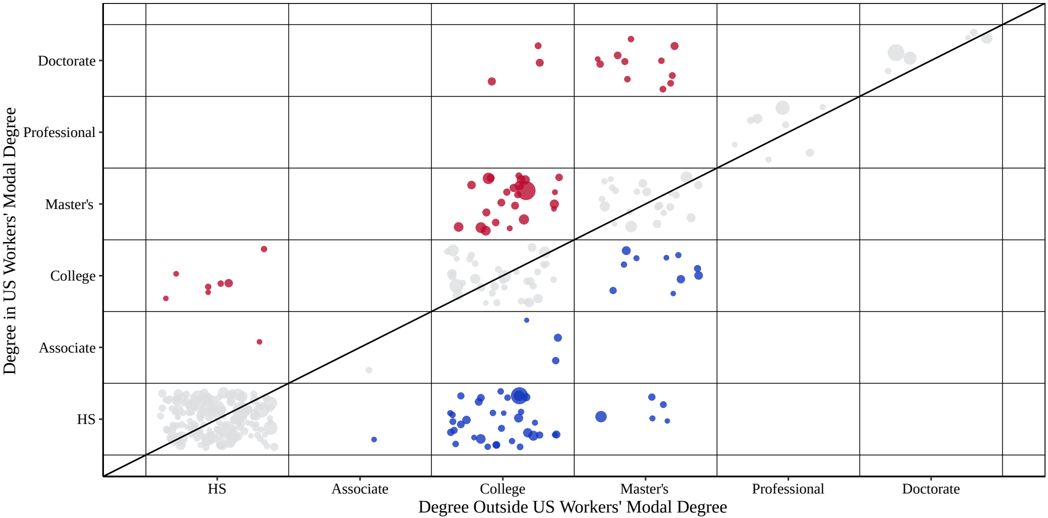
Figure A9: Educational Mismatch of Non-Citizens, 2010-2019



Location of Highest Degree Back

Figure A10: Educational Mismatch of Non-Citizens, Highest Degree in U.S., 2010-2019

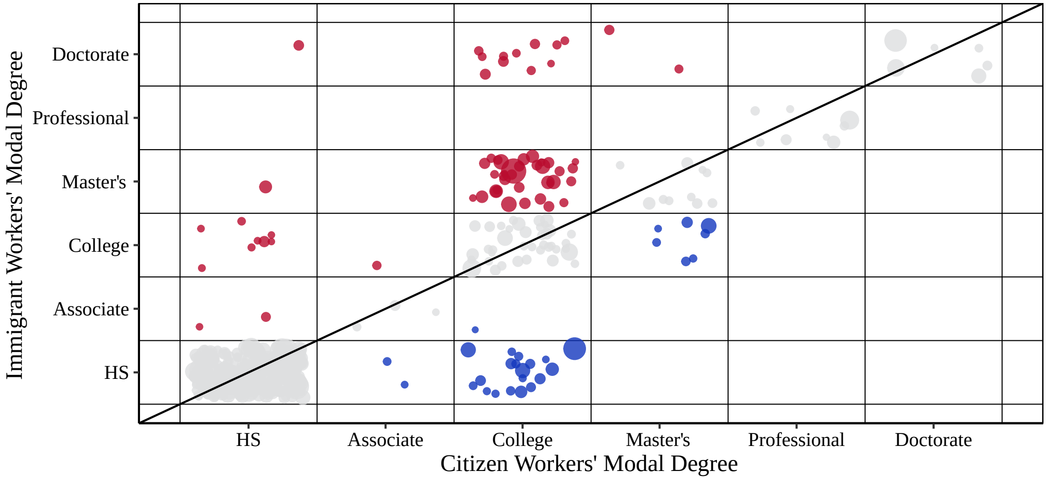


Figure A11: Educational Mismatch of Non-Citizens, Highest Degree Outside of U.S., 2010-2019

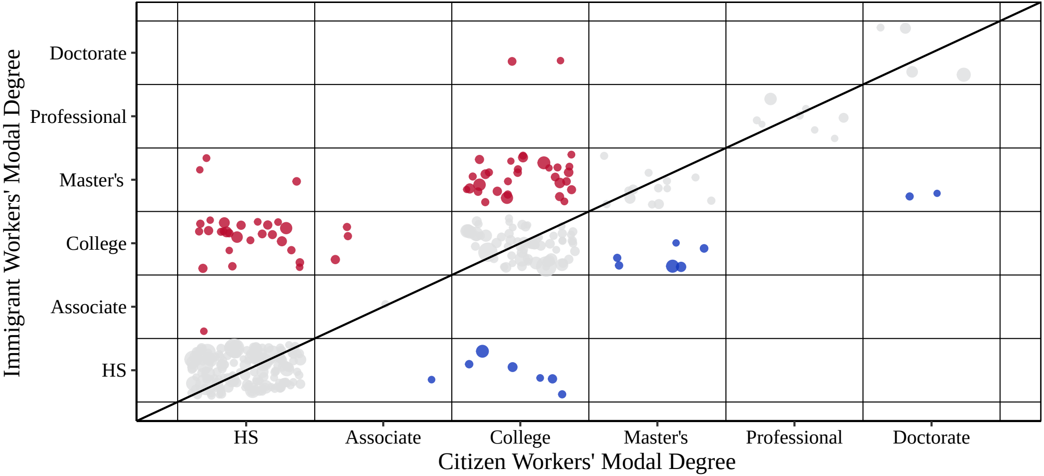
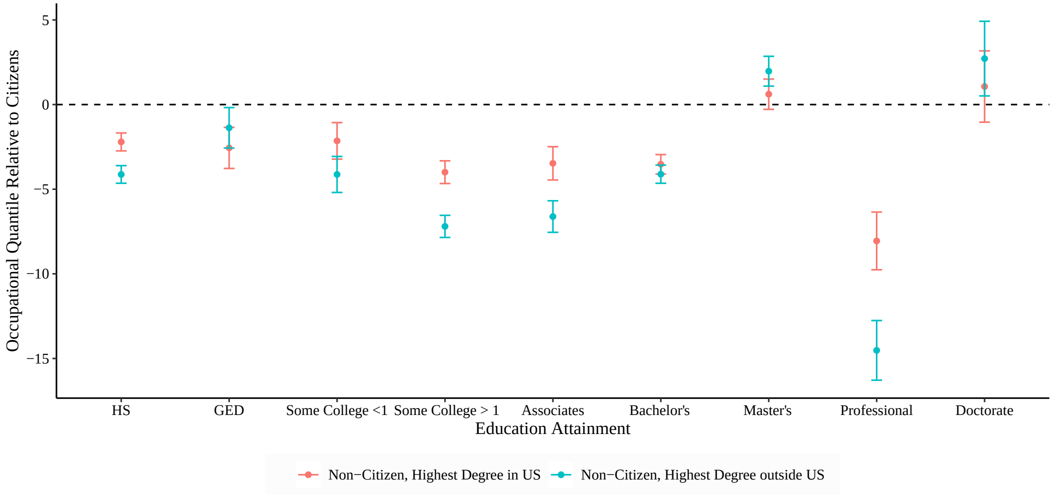


Figure A12: Estimated Deviations in Mean Occupational Quantile by Education Level for Immigrant Workers, 2010-2019 Back



Notes: 95% confidence intervals are shown in graph.